Humanoid Control Stack

build: passing | tests: 172/172 | license: CC BY 4.0 | python: 3.11

On-board control stack for a mobile surgical humanoid, gated by verify-before-generate.

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Index terms

About

This repository holds the humanoid control stack: the sensor-to-xyz pipeline, the planner, the broadcaster, and the safety checks. Every generated module clears the VVUQ gate before it is allowed to run.

Install and usage

```
$ pip install -e .
$ vvuq verify --interaction rpi.json
$ control-stack run --site PAT-NET-001
```

Repository structure

```
control-stack/
  sensor_to_xyz.py   10 Hz sensing
  planner.py         1 Hz LLM planner
  broadcaster.py     command fan-out
  xyz_safety.py      force and clearance checks
```

Figure placeholder (Images/topology.png). The float, caption, and \label mirror the VVUQ-02 figures; replace the box with \includegraphics.

Figure 1: Figure 1. Placeholder topology of the control stack modules.

Status

Component	Version	State
sensor_to_xyz	1.0.0	stable
planner	1.0.0	stable
xyz_safety	1.0.0	gated; 172/172

Table 1: Table 1. Component status (modeled on VVUQ-05 Table 1).

Acknowledgments

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Rights and Permissions

Cite This Manual

Kawchak K. Humanoid Control Stack Repository README. Zenodo. 2026; 10.5281/zenodo.xxxxxxxx.

Data Availability

Supporting records are openly available in the kevinkawchak/cancer-automated and kevinkawchak/physical-ai-oncology-trials GitHub repositories and are archived on Zenodo. All data sets are synthetic or illustrative and reproducible from a deterministic seed; no patient-identifiable information is included.

References

[1] K. Kawchak, "VVUQ Processing Priority over Code Generation: Method and Pipeline." https://doi.org/10.5281/zenodo.20372501, 2026. https://doi.org/10.5281/zenodo.20372501.

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